NYC 311 Service Requests Analysis Using Python

1. Project Overview

This project analyzes the **NYC 311 Service Requests** dataset using Python to identify patterns and trends in public service requests submitted by New York City residents. The analysis includes data cleaning, data preprocessing, exploratory data analysis (EDA), statistical analysis, feature engineering, and data visualization.

The objective of this project is to transform raw service request data into meaningful insights. The analysis helps identify the most common complaint types, evaluate agency performance, examine borough-wise request distribution, analyze request status, and measure service efficiency. These insights can support better decision-making, improve resource allocation, and enhance the overall management of public services.

2. Data Source

- **Dataset Name:** NYC 311 Service Requests
- **Source:** NYC Open Data Portal
- **File Type:** CSV
- **Number of Records:** 500,000
- **Number of Attributes:** 23

The dataset contains information about public service requests submitted by New York City residents. It includes details such as complaint type, responsible agency, borough, request status, creation and closure dates, reporting channel, and geographic coordinates. This information provides valuable insights into public service operations, complaint trends, and service request management across the city.

3. Problem Statement

New York City's 311 service receives a high volume of public service requests across multiple city departments every day. Analyzing these requests is essential to identify the most common public issues, understand service demand across different boroughs, evaluate agency workload, analyze reporting channels, and identify geographic hotspots.

Without proper analysis, it is difficult to recognize recurring complaint patterns, allocate resources effectively, and improve service efficiency. This project aims to analyze the NYC 311 Service Requests dataset using Python to uncover meaningful insights that support data-driven decision-making and enhance public service delivery.

4. Objectives:

1. Analyze NYC 311 service requests to identify major public service issues and complaint patterns.
2. Identify the most common complaint categories and compare their distribution across New York City boroughs.
3. Evaluate agency-wise service request volumes to understand department workload.
4. Analyze service request trends and request status to understand service demand.
5. Analyze citizen reporting channels to understand how service requests are submitted.
6. Identify complaint hotspots using geographic data (latitude and longitude) to locate high-demand service areas.

5. Attribute (Column / Feature) Details

Attribute Name	Description
unique_key	Unique identifier for each service request.
created_date	Date and time when the service request was created.
closed_date	Date and time when the service request was closed.
agency	Agency responsible for handling the request.
agency_name	Full name of the responsible agency.
complaint_type	Category of the reported complaint.
descriptor	Detailed description of the complaint.
location_type	Type of location where the complaint occurred.
city	City where the service request was submitted.
status	Current status of the service request.
resolution_description	Description of how the complaint was resolved.
resolution_action_updated_date	Date and time of the latest resolution update.
borough	Borough where the service request originated.
open_data_channel_type	Channel used to submit the request (e.g., phone, web, mobile app).
latitude	Latitude coordinate of the request location.
longitude	Longitude coordinate of the request location.

Attribute Name	Description
complaint_state	Engineered feature indicating whether the request is Open or Closed.
year	Engineered feature extracted from the request creation date.
month	Engineered feature representing the month of request creation.
month_name	Engineered feature showing the month name for easier interpretation.
day	Engineered feature representing the day of the month.
day_name	Engineered feature indicating the weekday of request creation.
hour	Engineered feature representing the hour when the request was created.

6. Tools & Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Plotly
- Google Colab
- Power BI
- n8n

7. Data Pre-Processing and Feature Engineering

Data Cleaning

The dataset was cleaned and prepared for analysis using the following data preprocessing steps:

- Performed Initial Exploratory Data Analysis (EDA) to examine the dataset structure, data types, missing values, duplicate records, and overall data quality.
- Checked the data types of all columns and verified the dataset structure.
- Identified missing values across the dataset.
- Retained missing values in the closed_date column because they represent service requests that are still open and have not yet been resolved.

- Created a new feature, complaint_state, from the closed_date column to classify each service request as Open or Closed.
- Imputed missing values in the descriptor column with "Unknown".
- Imputed missing values in the location_type column with "Unknown".
- Removed records with missing latitude or longitude values to ensure accurate geographic analysis.
- Removed the due_date column due to a high percentage of missing values and limited analytical value.
- Verified the dataset for duplicate records and removed them where necessary.
- Converted date-related columns into the appropriate datetime format.
- Performed Final Exploratory Data Analysis (EDA) on the cleaned dataset to validate preprocessing results and prepare the data for statistical analysis and visualization.

Feature Engineering

Feature engineering was performed to create additional features from the existing dataset to support more detailed analysis and improve the interpretation of service request patterns.

The following features were created:

- **Complaint State:** Created from the closed_date column to classify service requests as Open or Closed, making status analysis easier.
- **Year:** Extracted from the created_date column to enable year-wise analysis of service requests.
- **Month:** Extracted from the created_date column to analyze monthly service request trends.
- **Month Name:** Derived from the Month feature to improve the readability of visualizations and reports.
- **Day:** Extracted from the created_date column to analyze daily service request patterns.
- **Day Name:** Derived from the Day feature to identify service request trends across different days of the week.
- **Hour:** Extracted from the created_date column to analyze the hourly distribution of service requests and identify peak reporting hours.

These engineered features enhanced the dataset by enabling time-based analysis, improving data interpretation, and supporting more meaningful visualizations and insights.

8. Statistical Analysis

Measure of Central Tendency

Objective 4: Analyze service request trends and demand patterns.

The daily number of NYC 311 service requests was calculated by grouping the dataset based on the request creation date. Measures of central tendency were then computed to summarize the typical daily service request volume.

Statistical Results

- **Mean:** 10,414.09
- **Median:** 10,889
- **Mode:** 21

Chart Title

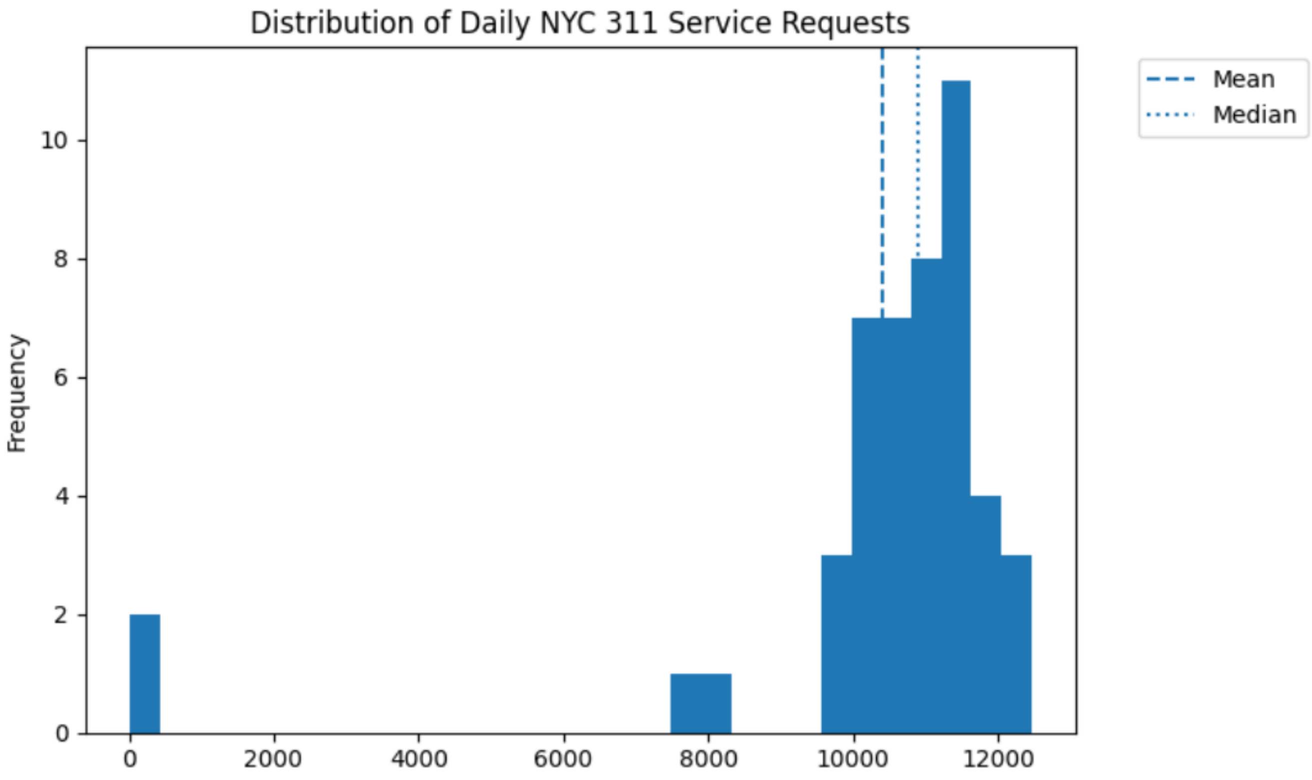
Distribution of Daily NYC 311 Service Request Counts

Summary / Interpretation

The histogram illustrates the distribution of daily NYC 311 service request counts over the analysis period.

- The **average (mean)** daily request count is **10,414**, indicating that the city receives approximately **10,000–11,000 service requests per day**.
- The **median** value of **10,889** is slightly higher than the mean, suggesting that most days recorded request volumes close to **11,000**.
- The **mode** is **21**, which represents the first day in the dataset containing only a partial day's data. Therefore, it should not be considered representative of the normal daily request volume.
- Most daily request counts are concentrated between **10,000 and 12,000**, indicating a relatively consistent demand for NYC 311 services throughout the observation period.
- The distribution also highlights a few unusually low request counts, which may be due to incomplete or partial-day data.

Figure 1:



Variance and Standard Deviation

Objective 4: Analyze service request trends and current request status to understand service demand.

Statistical Results

- **Variance: 5526419.08**
- **Standard Deviation: 2350.9**

Chart Title

Daily NYC 311 Service Request Variation

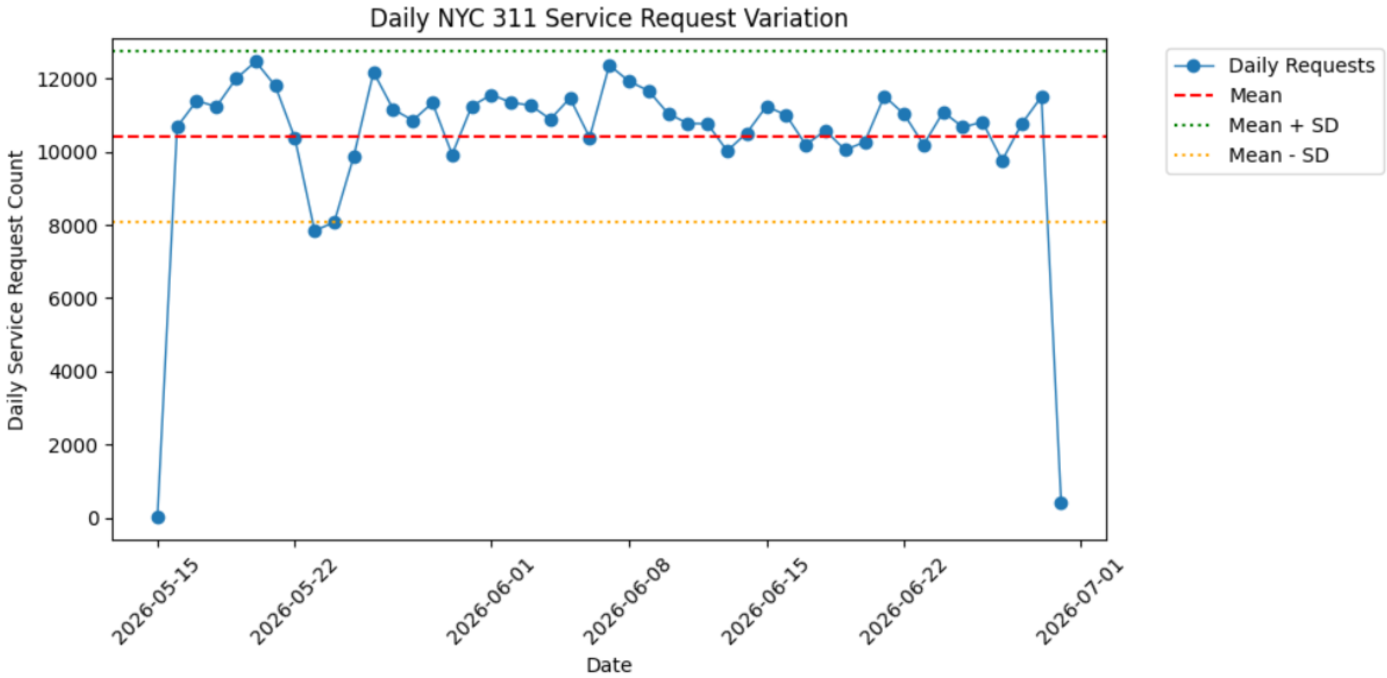
Summary / Interpretation

The line chart illustrates the daily variation in NYC 311 service request volumes over the analysis period. The red dashed line represents the average daily request count, while the green and orange dotted lines indicate one standard deviation above and below the mean, respectively.

- The variance measures how much the daily service request counts differ from the average, indicating the overall spread of the data.
- The standard deviation shows the typical variation in daily request volumes from the mean.
- Most daily request counts fall within one standard deviation of the mean, suggesting that service demand remained relatively stable throughout the analysis period.
- A few days fall outside this range, indicating unusually high or low service request volumes that may be influenced by external events, seasonal factors, or partial-day data.

- Overall, the chart shows that NYC 311 receives a consistently high number of daily service requests, with moderate fluctuations over time.

Figure 2:



Variance and Standard Deviation

Objective 3: Evaluate agency-wise service request volumes to understand department workload.

Statistical Results

- **Variance: 4,029,909,858.57**
- **Standard Deviation: 63,481.57**

Chart Title

Distribution of Agency-wise Service Request Volumes

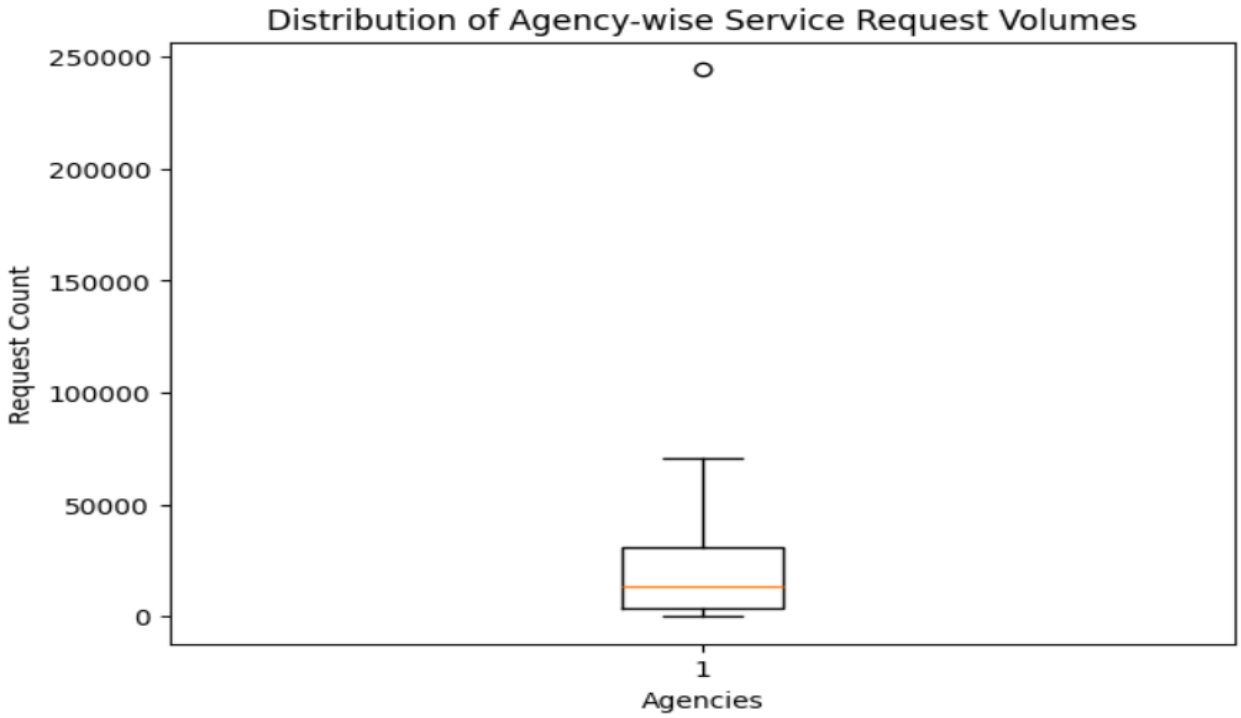
Summary / Interpretation

Variance and standard deviation were calculated to measure the variability in service request volumes across different NYC agencies.

- The **variance** of **4,029,909,858.57** indicates a substantial difference in the number of service requests handled by different agencies.
- The **standard deviation** of **63,481.57** shows that agency request volumes vary considerably from the average workload.
- The box plot illustrates that while most agencies receive **low to moderate numbers of service requests**, a few agencies handle significantly higher request volumes.
- The presence of an **outlier above the upper whisker** indicates that one agency receives an exceptionally high number of service requests compared to others.

- This uneven distribution suggests that service requests are concentrated among a small number of agencies, resulting in a heavier workload for those departments.
- These findings can help city administrators identify high-demand agencies and support resource planning, workload balancing, and service improvement initiatives.

Figure 3:



Skewness and Kurtosis

Objective 5: Analyze citizen reporting channels to understand how service requests are submitted.

Statistical Results

- **Skewness: 0.743**
- **Kurtosis: 1.716**

Chart Title

Distribution of Requests Across Reporting Channels

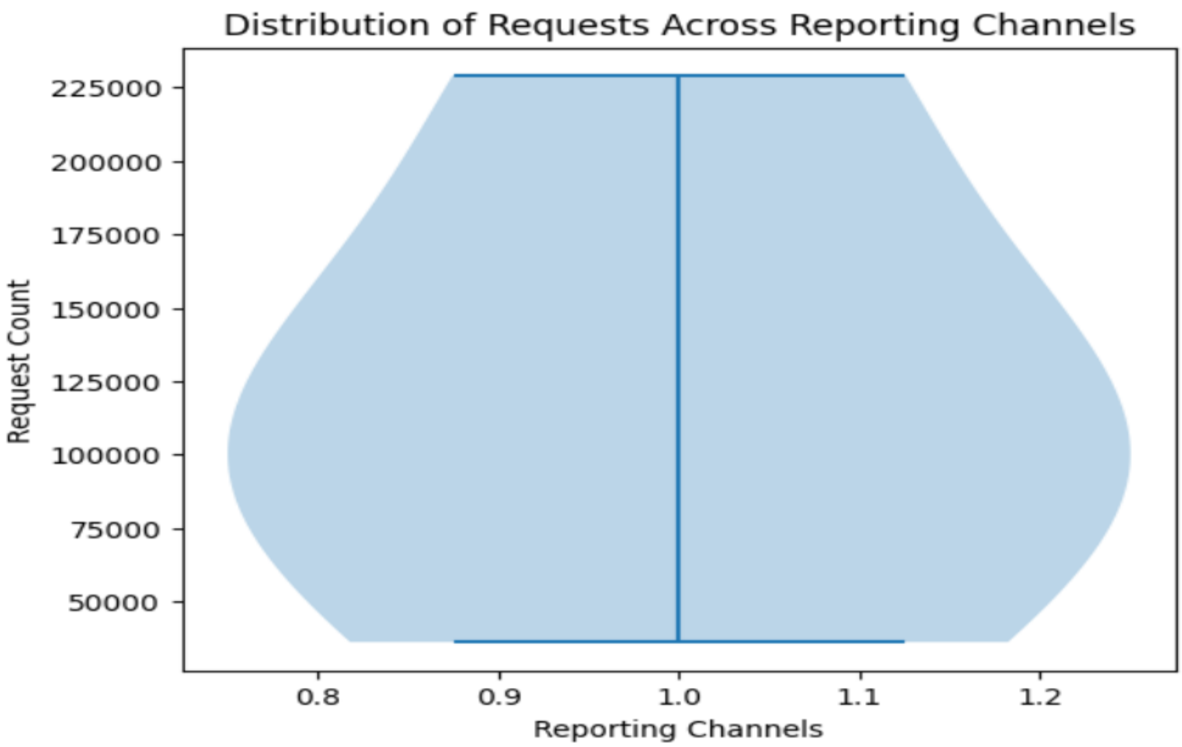
Summary / Interpretation

Skewness and kurtosis were calculated to understand the distribution of service requests across different reporting channels.

- The **skewness value of 0.743** indicates a **moderately positively skewed** distribution. This suggests that while most reporting channels receive relatively lower request volumes, a few channels receive significantly higher numbers of service requests.
- The **kurtosis value of 1.716** indicates a distribution with **slightly heavier tails than a normal distribution**, suggesting the presence of channels with unusually high request volumes.

- The violin plot shows that service request counts are not evenly distributed across reporting channels.
- The wider sections of the violin plot represent reporting channels with higher concentrations of service requests, while the narrower sections indicate channels with lower usage.
- These results suggest that citizens prefer certain reporting channels over others when submitting NYC 311 service requests.
- Understanding these usage patterns can help city administrators improve popular reporting channels, optimize resource allocation, and encourage the adoption of less frequently used channels.

Figure 4:



9. Data Visualization:

Univariate Analysis

1. Top 10 Most Reported Complaint Types

Objective 1: Analyze NYC 311 service requests to identify major public service issues and complaint patterns.

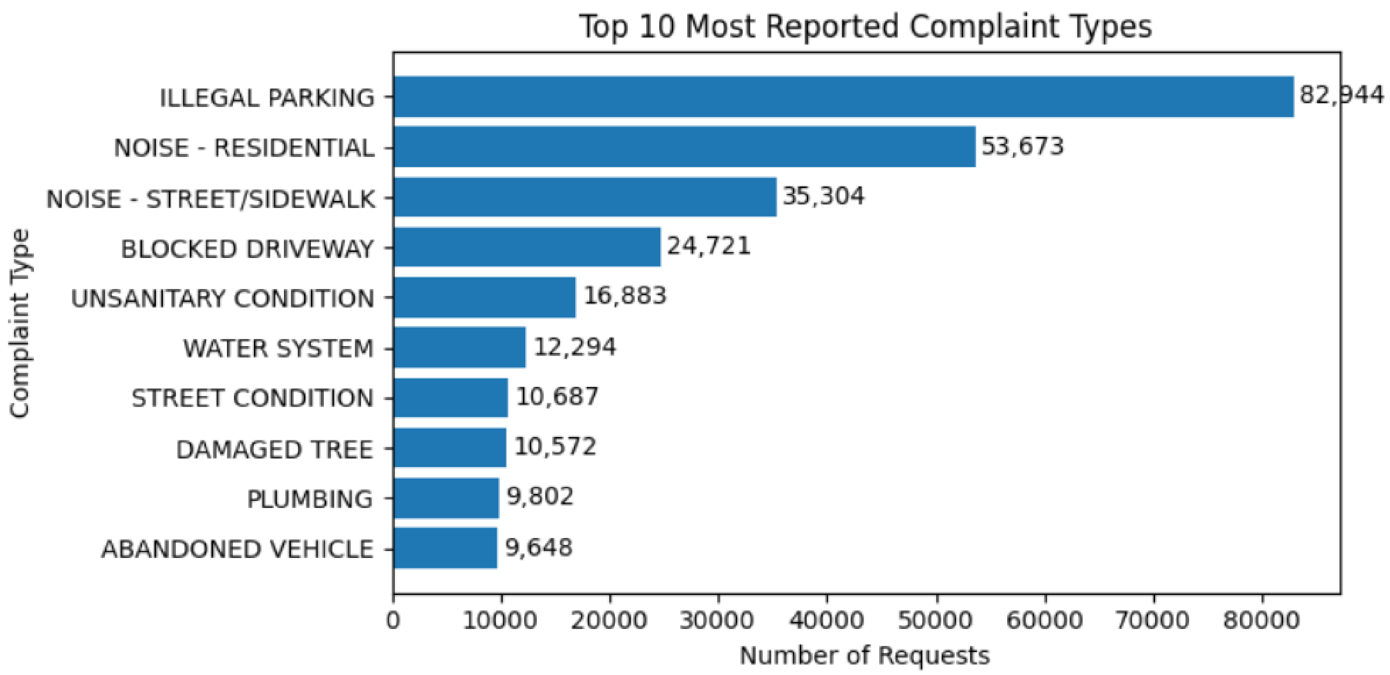
Summary / Interpretation

- The chart shows the **top 10 most reported complaint types** in the NYC 311 Service Requests dataset.
- **Illegal Parking** is the most reported complaint with **82,944** requests.
- **Noise – Residential** is the second most common complaint with **53,673** requests.
- **Noise – Street/Sidewalk** ranks third with **35,304** requests.
- **Blocked Driveway** and **Unsanitary Condition** are also among the frequently reported complaints.
- Overall, **parking and noise-related complaints** account for the highest number of service requests, indicating the major public service issues faced by New York City residents.

Business Insight

Key Insight: The analysis shows that parking and noise-related issues generate the highest number of 311 service requests. These insights can help city agencies prioritize resources and improve services in high-demand areas.

Figure 5:



2. Agency-wise Service Request Volumes

Objective 3: Evaluate agency-wise service request volumes to understand department workload.

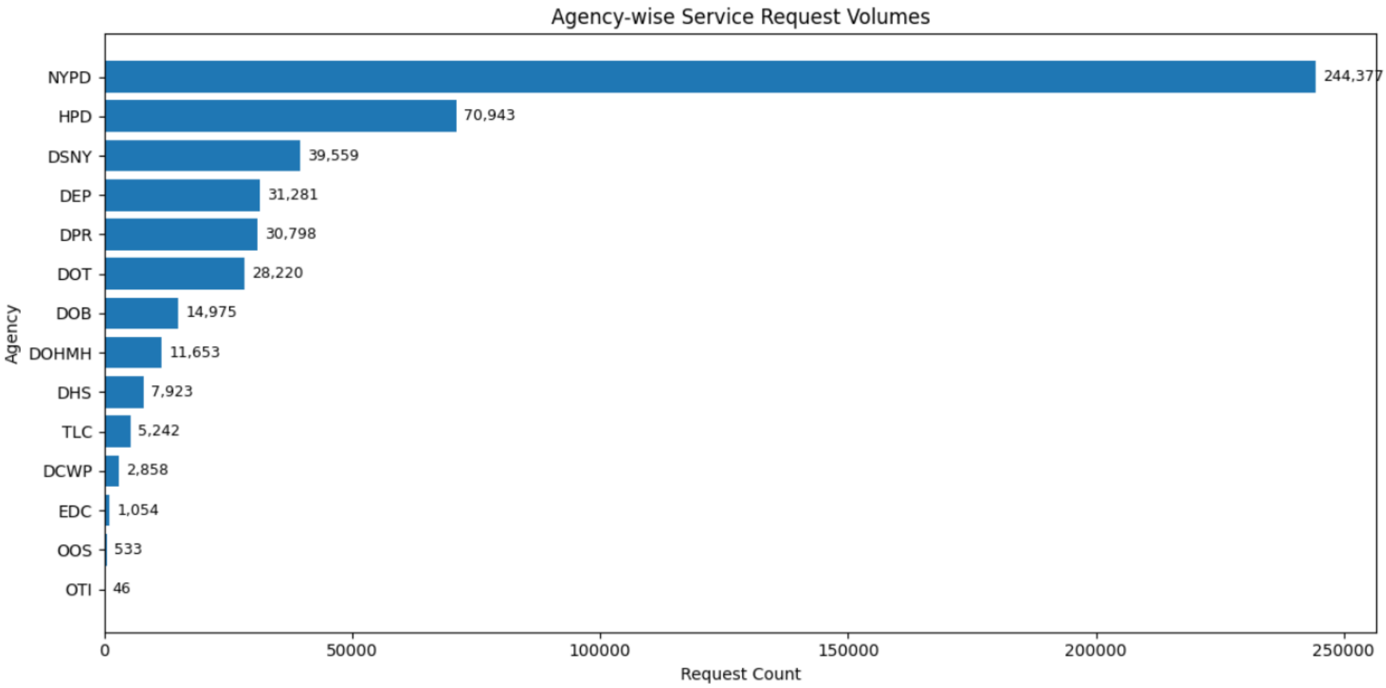
Summary / Interpretation

- The chart shows the number of service requests handled by different NYC agencies.
- **NYPD** received the highest number of service requests (**244,377**), indicating it has the largest workload among all agencies.
- **HPD (Housing Preservation and Development)** is the second busiest agency with **70,943** requests, followed by **DSNY (Department of Sanitation)** with **39,559** requests.
- **DEP, DPR, and DOT** also receive a considerable number of service requests, reflecting steady demand for their services.
- Agencies such as **DOB, DOHMH, DHS, TLC, DCWP, EDC, OOS, and OTI** handle comparatively fewer service requests.
- Overall, the chart indicates that **service requests are concentrated among a few key agencies**, while the remaining agencies receive a much lower volume of requests.

Business Insight

Key Insight: The analysis shows that NYPD, HPD, and DSNY handle the majority of NYC 311 service requests. Understanding agency-wise workload can help city administrators allocate resources efficiently and improve service delivery in high-demand departments.

Figure 6:



3. Distribution of Service Request Status

Objective 4: Analyze service request trends and current request status to understand service demand.

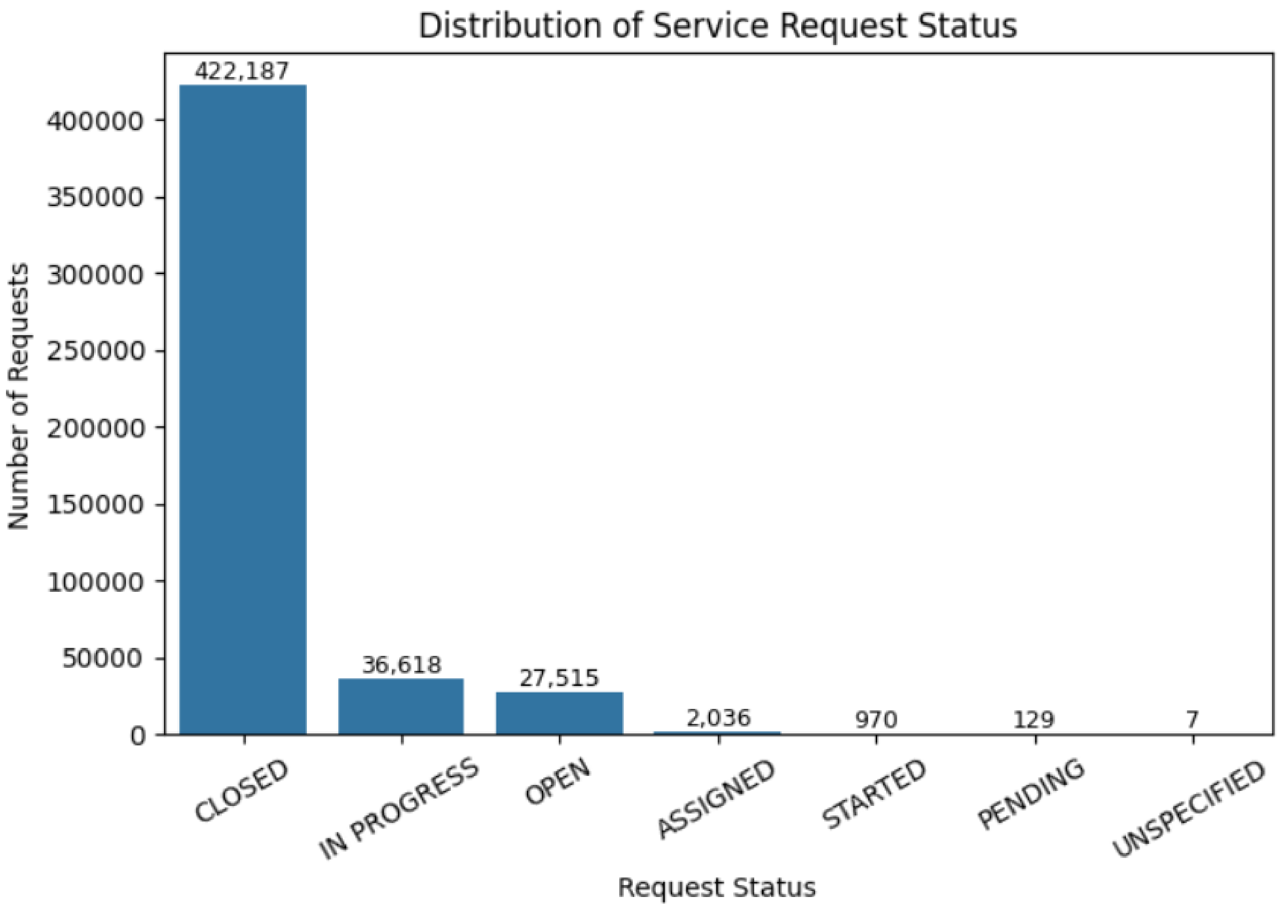
Summary / Interpretation

- The chart shows the distribution of NYC 311 service requests based on their current status.
- **Closed** is the most common status, with **422,187** requests, indicating that the majority of service requests have been resolved.
- **In Progress** has **36,618** requests, showing that some requests are currently being worked on.
- **Open** accounts for **27,515** requests, representing requests that are yet to be resolved.
- **Assigned, Started, Pending,** and **Unspecified** make up only a small percentage of the total requests.
- Overall, the chart indicates that **most NYC 311 service requests are successfully closed**, while only a small proportion remain in progress or open.

Business Insight

Key Insight: The high number of closed requests suggests effective service resolution by NYC agencies. Monitoring the remaining open and in-progress requests can help improve response times and overall service efficiency.

Figure 7:



4. **Service Requests by Reporting Channel**

Objective 5: Analyze citizen reporting channels to understand how service requests are submitted.

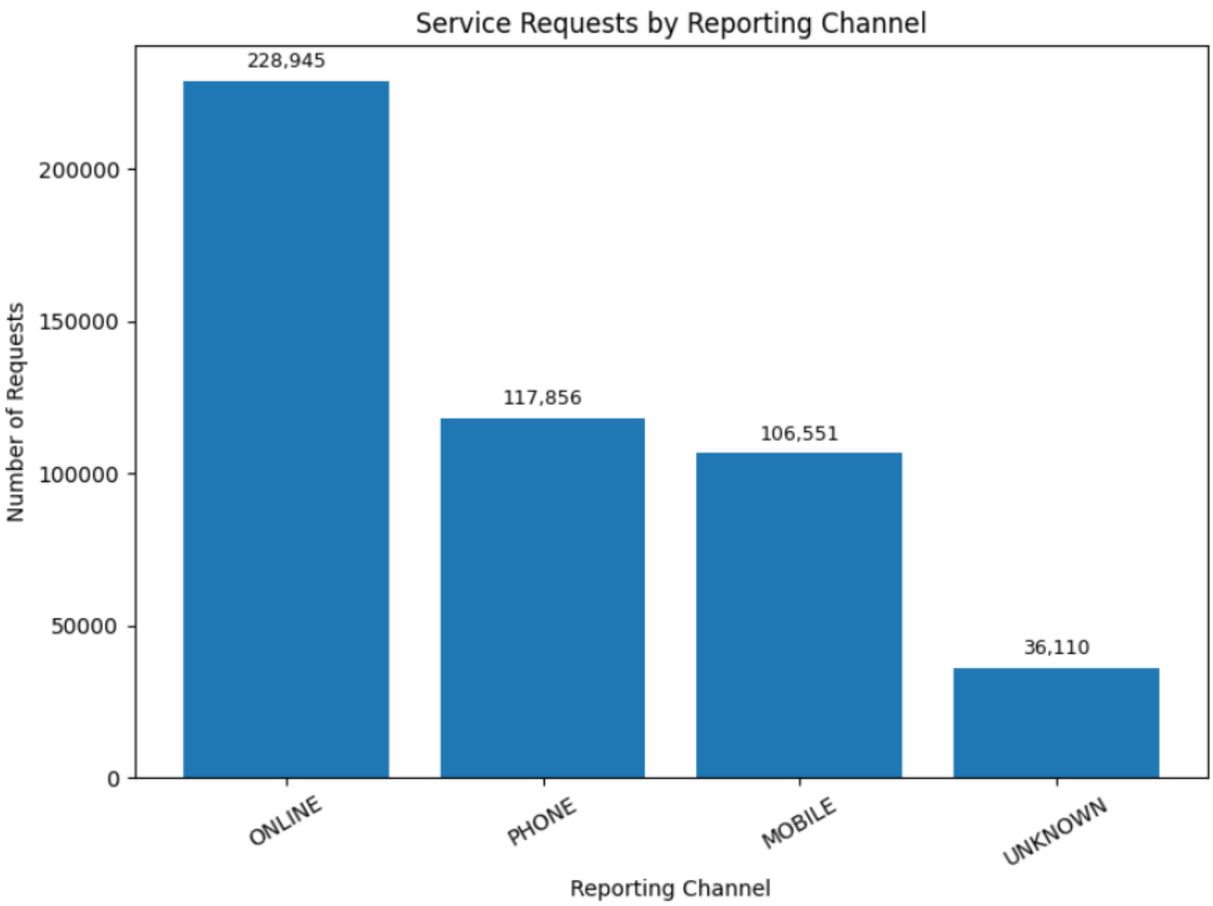
Summary / Interpretation

- The chart shows the distribution of NYC 311 service requests across different reporting channels.
- **Online** is the most frequently used reporting channel, with **228,945** service requests.
- **Phone** is the second most used reporting channel, with **117,856** requests.
- **Mobile** follows closely with **106,551** service requests.
- **Unknown** has the lowest number of requests, with **36,110**.
- Overall, the results indicate that **online reporting is the preferred method for submitting NYC 311 service requests**, while phone and mobile channels are also widely used.

Business Insight

Key Insight: Citizens primarily use online platforms to submit service requests. This insight can help city agencies focus on improving digital services while ensuring phone and mobile channels continue to support residents effectively.

Figure 8:



Bivariate Analysis

5. Complaint Categories Across NYC Boroughs

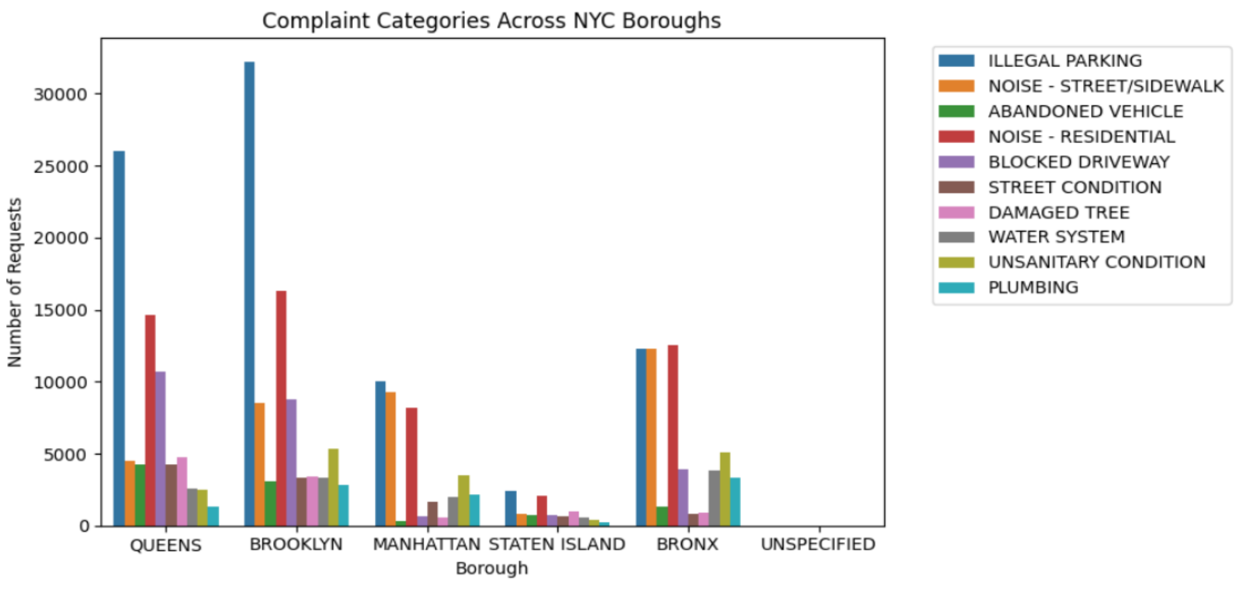
Objective 2: Identify the most common complaint categories and compare their distribution across NYC boroughs.

- **Illegal Parking** is the most common complaint category across all NYC boroughs, with **Brooklyn and Queens recording the highest number of requests**, indicating significant parking-related issues in these areas.
- **Noise complaints** are also major concerns:
    - **Noise – Residential** shows high volumes, especially in **Bronx, Brooklyn, and Queens**.
    - **Noise – Street/Sidewalk** is more prominent in **Brooklyn and Bronx**, highlighting public disturbance issues.
- **Blocked Driveway complaints** are higher in **Queens and Brooklyn**, suggesting challenges related to vehicle access and parking violations.
- **Unsanitary Condition complaints** appear notably higher in **Brooklyn and Bronx**, indicating increased sanitation-related service needs.
- **Manhattan shows comparatively lower complaint volumes** across most categories except for major issues like illegal parking and noise-related complaints.
- **Staten Island has the lowest overall complaint volume** among the boroughs, while **Brooklyn and Queens experience the highest demand for city services**.

Key Insight:

The analysis highlights that **parking violations, noise disturbances, and sanitation issues are the dominant public service concerns in NYC**, with **Brooklyn and Queens requiring greater attention and resource allocation due to higher complaint volumes**.

Figure 9:



6. **Daily Service Request Trend**

Objective 4: Analyze service request trends and current request status to understand service demand.

Summary / Interpretation

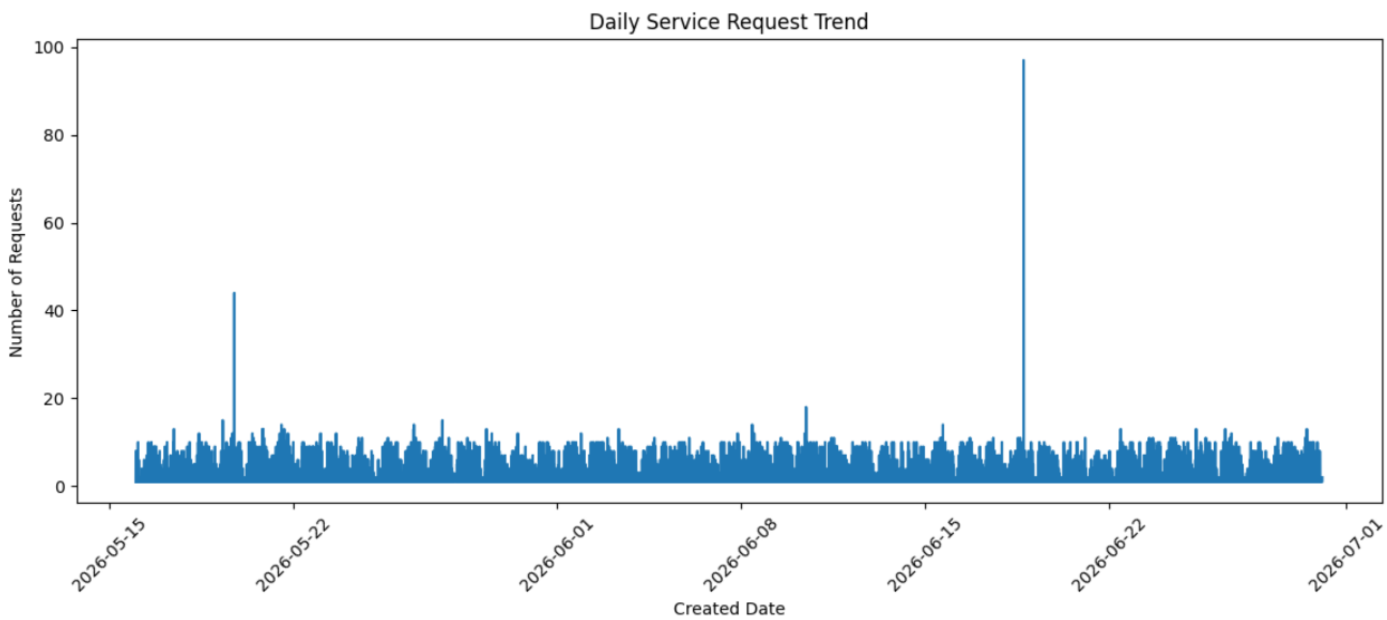
The chart shows the daily trend of NYC 311 service requests during the analysis period.

- The number of daily requests remains mostly consistent, showing a steady demand for NYC 311 services.
- A few days show higher request volumes, indicating periods of increased service demand.
- The highest peak is observed around **mid-June 2026**, where requests increased significantly.
- Another small increase is seen during **late May 2026**.
- Overall, the trend shows a stable flow of service requests with occasional spikes.

Business Insight

- A consistent request pattern helps NYC agencies plan regular staffing and daily operations.
- High-demand periods require better resource allocation to maintain faster response times.
- Monitoring request trends helps agencies identify demand patterns and improve service efficiency.

Figure 10:



7. Geographic Distribution of NYC 311 Service Requests

Objective 6: Identify complaint hotspots using geographic data (latitude and longitude) to locate high-demand service areas.

Summary / Interpretation

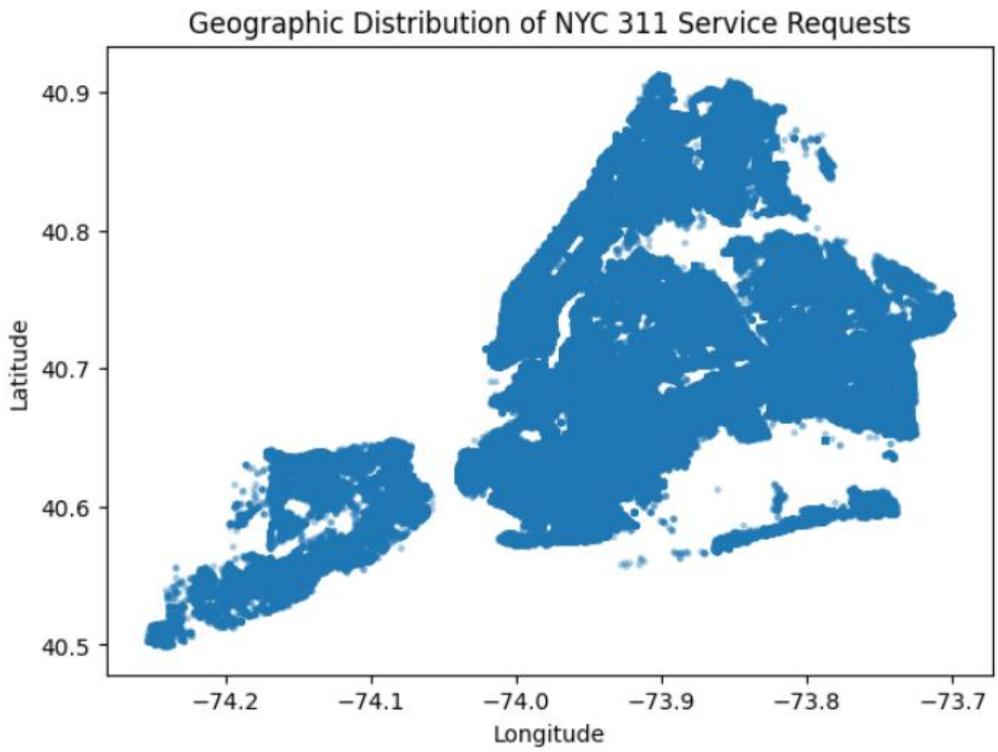
The map shows the geographic distribution of NYC 311 service requests using latitude and longitude data.

- Service requests are spread across all five NYC boroughs, showing that citizens actively use the 311 system throughout the city.
- Higher concentrations of requests are visible in areas such as **Brooklyn, Queens, Bronx, and Manhattan**, indicating greater service demand in these regions.
- **Staten Island** shows comparatively fewer requests than other boroughs.
- The clustering of request locations highlights specific complaint hotspots where service issues are reported more frequently.

Business Insight

- Identifying complaint hotspots helps NYC agencies prioritize high-demand areas and allocate resources effectively.
- Geographic analysis supports better planning of field operations and improves response times.
- Understanding location-based complaint patterns helps agencies address recurring issues and improve public service delivery.

Figure 11:



Multivariate Analysis

8. Top 10 Complaint Categories by Borough

Objective 2: Identify the most common complaint categories and compare their distribution across NYC boroughs.

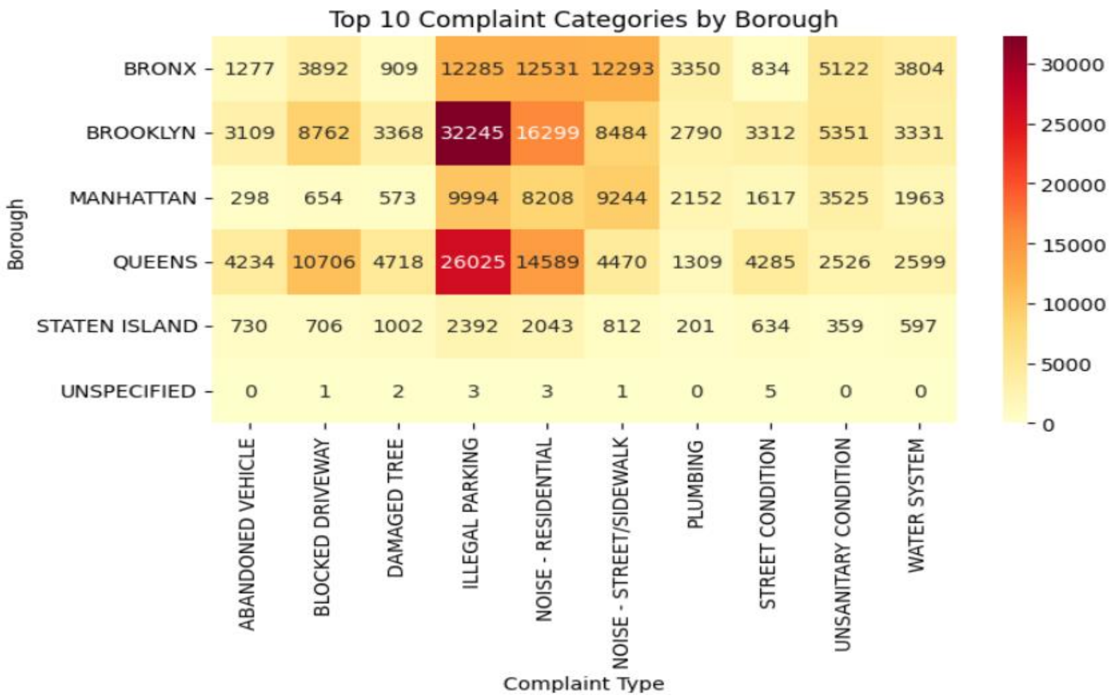
Summary / Interpretation:

- **Brooklyn** has the highest number of service requests among all boroughs, with **Illegal Parking** being the most common complaint category.
- **Queens** also shows a high volume of complaints, where **Illegal Parking** is the leading issue reported by residents.
- **Bronx** has significant complaints related to **Noise - Residential** and **Illegal Parking**.
- **Manhattan** shows a considerable number of complaints, mainly related to **Illegal Parking** and **Noise - Street/Sidewalk**.
- **Staten Island** has comparatively fewer complaints across all major categories.
- Overall, **Illegal Parking** is the most frequently reported complaint type across NYC, followed by **Noise - Residential** and **Noise - Street/Sidewalk**.

Business Insight

- **Brooklyn and Queens** require more attention due to their higher complaint volumes.
- Parking and noise-related issues are the major service concerns across NYC and should be prioritized for better enforcement and resolution.
- Borough-level complaint analysis helps agencies allocate resources effectively and improve public service response.

Figure 12:



9. **Agency-wise Complaint Distribution Across Boroughs**

Objective 3: Evaluate agency-wise service request volumes to understand department workload.

Summary / Interpretation:

- **NYPD** handles the highest number of service requests across NYC boroughs, indicating the largest operational workload.
- **HPD** receives the next highest volume of requests, mainly related to housing and property service issues.
- Other agencies such as **DSNY, DOT, DPR, and DEP** also manage a significant number of service requests across different boroughs.
- **Brooklyn and Queens** show higher service request volumes for most agencies, indicating greater demand for public services.
- **Staten Island** records comparatively fewer requests across agencies.
- Agencies such as **OTI, OOS, EDC, and DCWP** handle fewer requests compared to major service agencies.

Business Insight

- High-demand agencies like **NYPD and HPD** require effective workload management and resource planning.
- Boroughs with higher request volumes need better staff allocation and operational support.
- Agency-wise analysis helps NYC departments identify workload patterns, improve response efficiency, and enhance service delivery.

Figure 13:

borough	BRONX	BROOKLYN	MANHATTAN	QUEENS	STATEN ISLAND	UNSPECIFIED
agency						
DCWP	468	845	816	646	82	1
DEP	5795	9686	6078	8366	1355	1
DHS	394	1516	5312	641	59	1
DOB	2627	5113	2639	3851	744	1
DOHMH	1719	3748	2991	2631	562	2
DOT	3057	8639	5456	8956	2106	6
DPR	2972	10492	3409	11334	2588	3
DSNY	5501	13341	7451	10419	2844	3
EDC	14	253	651	116	20	0
HPD	22592	21967	15872	9263	1249	0
NYPD	48021	78484	39492	70631	7739	10
OOS	102	124	167	123	17	0
OTI	7	10	21	8	0	0
TLC	169	716	3223	1114	19	1

10. Summary of Findings

- **Illegal Parking** was identified as the most common complaint category, followed by **Noise - Residential** and **Noise - Street/Sidewalk**.
- **Brooklyn and Queens** recorded the highest number of service requests, indicating greater public service demand in these boroughs.
- **NYPD** handled the highest volume of requests, followed by **HPD**, showing higher workload among these agencies.
- Most service requests were marked as **Closed**, indicating effective resolution of reported issues.
- Daily service request trends showed a **stable demand pattern** with occasional spikes during high-demand periods.
- Geographic analysis identified complaint hotspots across NYC, helping locate areas with higher service needs.
- The analysis supports better **resource allocation, workload management, and faster response planning** for NYC agencies.

11. Advanced Analytics Insights

1. Descriptive Analysis

- Identified the most common complaint categories.
- Compared complaint volumes across NYC boroughs.
- Analyzed agency-wise service request volumes.
- Reviewed the distribution of service request status (Open, Closed, In Progress, etc.).
- Examined the reporting channels used by citizens.
- Identified complaint hotspots using geographic (latitude and longitude) data.

2. Diagnostic Analysis

- Identified why certain complaint categories have higher request volumes.
- Analyzed complaint distribution across different boroughs.
- Determined which agencies handle the highest workload.
- Examined geographic areas with recurring complaint hotspots.
- Analyzed citizens' preferred channels for reporting service requests.

3. Predictive Analysis

- High-frequency complaint categories, such as **Illegal Parking** and **Noise Complaints**, are likely to continue generating large numbers of requests.
- **Brooklyn and Queens** are expected to remain high-demand areas based on historical trends.
- Agencies such as **NYPD** and **HPD** may continue to experience higher workloads.
- Existing complaint hotspots are likely to generate future service requests unless preventive measures are implemented.

4. Prescriptive Analysis

- Allocate additional resources to high-demand boroughs.
- Prioritize recurring complaint categories to reduce future service requests.
- Improve workforce planning for agencies with higher workloads.
- Increase inspections and preventive actions in complaint hotspot locations.
- Strengthen digital reporting systems and monitor complaint trends regularly to improve service delivery.

12. Project Extension – Automation Workflow

Overview

As an extension of the **NYC 311 Service Requests Analysis project**, an end-to-end automation workflow was developed using **n8n, Python, Google Sheets, and Power BI** to automate data processing, reporting, and dashboard updates.

Key Features

- Automated data extraction and processing.
- Automated data updates to Google Sheets.
- Automated dashboard data refresh.
- Reduced manual intervention in the reporting process.
- Improved data accuracy and reporting efficiency.
- Ensured up-to-date insights for analysis and decision-making.

Tools Used

- n8n
- Python
- Google Sheets
- Power BI

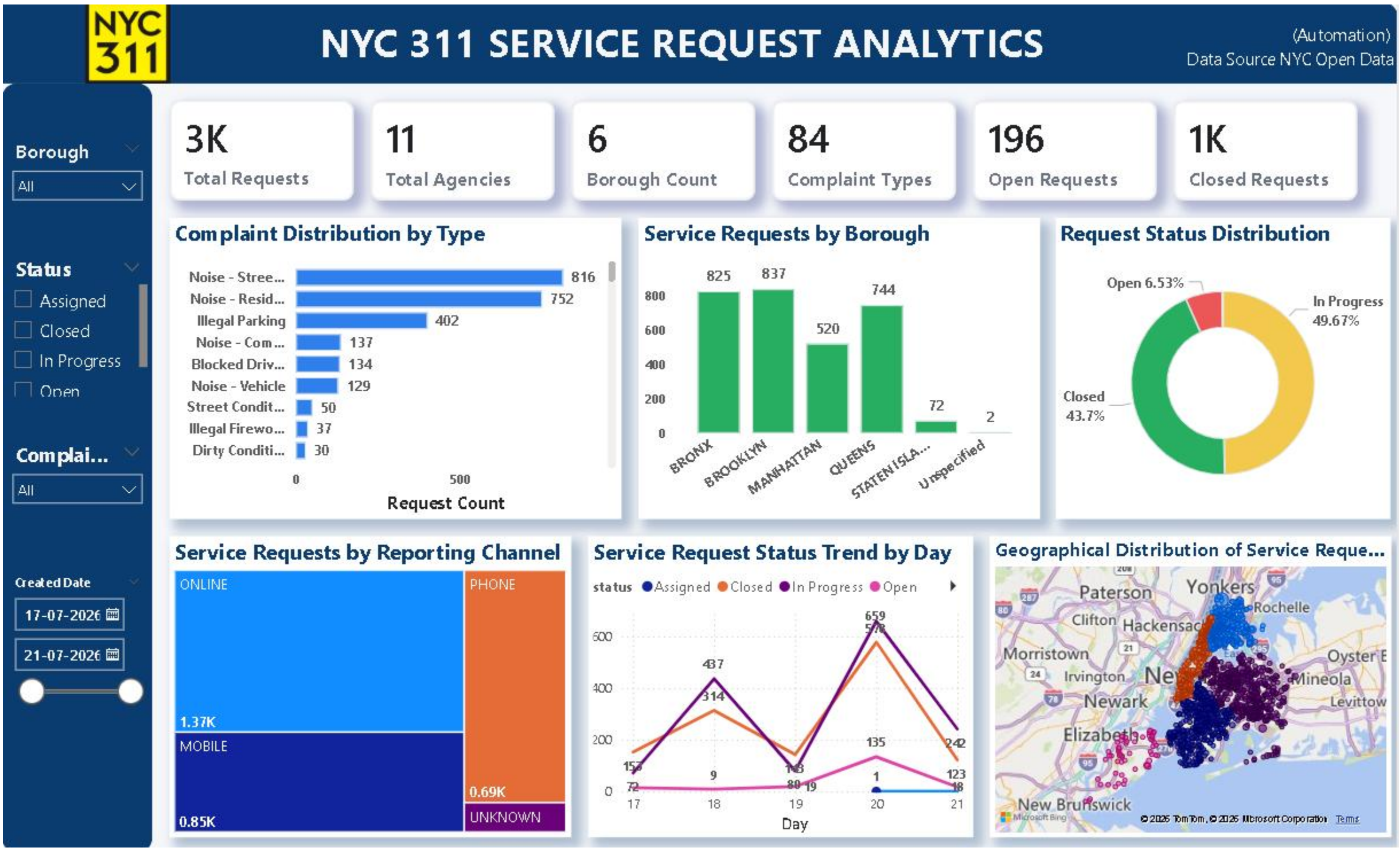
Project Repository

For the complete automation workflow, implementation details, source code, and documentation, please refer to the GitHub repository:

GitHub Repository:
NYC 311 Service Requests Analytics Automation Pipeline (n8n + Python + Power BI)
https://github.com/ShaliniSakthivel-DA/NYC-311-Service-Requests-Analytics-Automation-Pipeline-n8n-Python-PowerBI

13. PowerBI Dashboard:

Figure 14:



An interactive **Power BI dashboard** was developed to visualize key insights from the NYC 311 Service Requests dataset. The dashboard provides an easy-to-understand overview of complaint patterns, service demand, agency workload, and geographic distribution, enabling users to monitor public service performance and support data-driven decision-making.

Dashboard Components

KPI Cards

The dashboard includes KPI cards to provide a quick summary of the overall service request data:

- **Total Service Requests** – Displays the total number of NYC 311 service requests analyzed.
- **Total Complaint Categories** – Shows the number of unique complaint types reported.
- **Total Agencies** – Displays the number of agencies responsible for handling service requests.

- **Total Boroughs** – Shows the number of NYC boroughs included in the analysis.

Dashboard Visualizations

The dashboard includes the following interactive visualizations:

- **Top 10 Complaint Types** – Identifies the most frequently reported complaint categories.
- **Agency-wise Service Request Volumes** – Compares the workload handled by different NYC agencies.
- **Service Request Status Distribution** – Displays the current status of service requests (Closed, Open, In Progress, etc.).
- **Reporting Channel Analysis** – Shows how citizens submitted their service requests (Online, Phone, Mobile, etc.).
- **Daily Service Request Trend** – Illustrates changes in daily service request volumes over time.
- **Complaint Categories by Borough** – Compares major complaint categories across NYC boroughs.
- **Agency-wise Complaint Distribution Across Boroughs** – Evaluates agency workload by borough.
- **Geographic Complaint Hotspots** – Maps service request locations to identify high-demand areas across NYC.

14. Conclusion

The **NYC 311 Service Requests Analysis** project successfully transformed raw public service request data into meaningful insights using **Python and Power BI**. Through data cleaning, feature engineering, exploratory data analysis (EDA), statistical analysis, and data visualization, the project identified complaint patterns, agency workload, borough-wise distributions, service request trends, reporting channels, and complaint hotspots.

The insights from this project can help city agencies improve resource allocation, enhance response times, and support better decision-making. In addition, an **automation workflow** was developed to simplify data processing and reporting, making the overall analysis more efficient.

Overall, this project demonstrates practical skills in **data analysis, data visualization, dashboard development, and workflow automation**, providing a strong foundation for real-world data analytics projects.